Review: The ALeRCE text-to-SQL system

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Summary

The paper builds a natural-language query layer over the ALeRCE broker database (25 tables, 304 columns; 27,000+ users in 139 countries) using in-context learning only. Following DIN-SQL and DAIL-SQL, the pipeline is schema linking, difficulty classification, decomposition into a step-by-step (sbs) plan for medium and hard queries, and one self-correction pass fired on an execution error. Evaluation rests on 110 NL/SQL pairs — 21 from ALeRCE workshops, 89 questions from 10 astronomers, one SQL expert authoring every gold query — split 58 dev / 52 test (32 simple, 10 medium, 10 hard). The metric is an asymmetric perfect match (PM), binary per query and built on set precision/recall: PM_rows fires only at precision = recall = 1 on returned row identifiers, while PM_cols requires only recall = 1, forgiving extra columns. Thirteen models run ten times each are compared with unpaired permutation tests (Holm-corrected, $\alpha = 0.05$), plus per-stage dollar accounting.

Main contributions

1. **A domain-specific astronomical T2S gold set that is not Spider.** The public 110-pair set is grounded in real broker traffic, including deliberately rare queries against `dataquality`, PS1, WISE and Solar-System tables (Sect. 3.1). ScienceBenchmark showed Spider-tuned methods at >80% execution accuracy dropping to 21% on SDSS.
2. **A decomposed pipeline that beats direct inference, with two of its four modules ablated.** For Opus 4.6, sbs beats direct significantly on simple and hard queries (a draw on medium); sbs outranks direct for five of six models in Table 2 and holds six of the top seven positions. The exception, Gemini 3 Flash, ranks third direct at SUM 13 against its own sbs run's 14.
3. **Self-correction is the cheapest win.** It improves or ties 154 of the 156 w/o-vs-w/ comparisons in Table D.1 — the only regressions are Claude 3.7's hard-column PM (0.31 to 0.28 direct, 0.28 to 0.26 sbs); no top-six model loses — and lifts Opus 4.6 simple-row PM from 0.844 to 0.938 (direct) and 0.938 to 0.969 (sbs) for $0.09 of the $2.87 sbs run. Fig. 9 shows all but one non-timeout error repaired; the residue is ALeRCE's 2-minute timeout.
4. **First-class cost accounting.** Table 3 splits dollars per stage: Opus 4.6 sbs at $2.87 per test-set run is 2.41x Gemini 2.5 Pro sbs, decomposition ($1.07) and generation ($1.05) dominating, while Gemini 3 Flash sbs ties for fourth on SUM (14, with GPT-5.2-Codex sbs) at $0.27 — a tenth of the price. Table E.1's token budget covers twelve of the thirteen models (Gemini 2.5 Flash is absent).
5. **Negative results are reported, not buried.** Masked-question-similarity 3-shot lifts Gemini 2.5 Pro (direct) on the medium bin by 18 points on rows (0.42 to 0.60) and 14 on columns (0.72 to 0.86), with smaller gains on simple and hard (Table 5). Yet of the six few-shot settings in Table F.2, five significantly depress Opus 4.6 sbs hard rows from 0.59 (to 0.38 at 3-shot Random) and none improves them; structured outputs significantly degrade medium rows for both models tested (Gemini 3 Flash 0.40 to 0.30, Opus 4.6 0.44 to 0.30; Table G.2).

Critical assessment

The statistics are the weakest link. Medium and hard each hold **ten test queries**, so a single run's PM there can only be a multiple of 0.1; averaging ten runs only re-quantises the mean to 0.01, and the near-deterministic models (Opus 4.6, Gemini 3 Flash, Sonnet 4.5) land back on bare multiples of 0.1 with sd 0 or 5.9e-17. The quantum that matters is one query. Variance is estimated across runs, not queries, so the permutation tests measure noise the pipeline barely has and ignore item sampling: a "significant" 0.1 gap on hard rows is one query changing state. The abstract's non-monotonicity — Opus 4.6 sbs at 0.59 hard rows against 0.44 medium, i.e. 5.9 versus 4.4 queries — inherits this and goes undiagnosed. Difficulty is set by table and sub-query *counts* (Sect. 3.2), not measured error, so the taxonomy driving prompt routing is unvalidated. Bootstrapping over queries would fix it for free.

Second, two of the four modules are never ablated. Direct-versus-sbs measures decomposition and w/-versus-w/o self-correction, but with no oracle-schema and no fixed-difficulty condition the paper cannot say how much of the residual error — 0.44 medium, 0.59 hard rows — is schema linking, which their own Sect. 2 cites Li et al. (2024) as finding the largest T2S error class on BIRD at 41.6%. Both are one prompt change away.

Third, the gold set has one author and no ambiguity budget. Sect. 2.1 names NL ambiguity as a core T2S challenge, yet a defensible alternative reading scores a hard zero, since PM_rows demands exact set equality; with no second SQL expert, 0.44 on medium could be a model failure or one annotator's conventions. Those conventions leak into the system under test: the timeout prompt (Listing C.3) tells the model to set `ranking=1` whenever `probability` is joined "unless the request said otherwise," and to "ensure there are at least 3 conditions on the probability table" — gold-query idioms injected at test time, on the error class Fig. 9 calls dominant, i.e. the queries that decide the hard-row score. Stripping both clauses and re-running the ten hard queries would settle it.

Fourth, this is prompt engineering, not an agent: the pipeline runs once and re-prompts only on an execution *error*, never checking returned rows for plausibility or revising a query that executes but answers the wrong question — a limit the paper honestly flags, noting that semantic errors survive self-correction (Fig. 10). The sweep is closed-weights only, and the observation that newer is not better — GPT-5.3-Codex below GPT-5.2-Codex, SUM 15 versus 14 for sbs — is unexplained; the apparent second instance is a naming artefact, since Table 1 shows "Gemini 3.1 Flash" is `gemini-3.1-flash-lite-preview`, a tier difference, not a generation one. One inconsistency needs fixing: Opus 4.6 sbs hard-column PM is 0.31 in Tables D.1 and G.2 but 0.49 in Table F.2 and 0.48 in G.1, with the abstract quoting 0.49.

Relation to the agentic lensing programme

(1) Labels are the wall, again. One SQL expert wrote all 110 gold queries, including the 89 questions from astronomers who never saw SQL, so the 0.44 medium and 0.59 hard row PMs have no measured ceiling. We ran that experiment on the imaging side; the ceiling was the result: the Huang Paper-II two-grader recovery gives 55.4% exact agreement, symmetric QWK 0.42, grader-vs-consensus QWK 0.776, while identical engineered features score AUC $\sim$0.80 on spectroscopically confirmed labels and $\sim$0.51 on the hard consensus pool. Inter-team ceilings are lower still (DESI vs SuGOHI QWK 0.29, n=103; DESI vs Euclid panel 0.17, n=24). A 20-query second-expert replication of their gold SQL would bound PM as our 250-item, 40-repeat single-grader campaign is designed to bound QWK.

(2) Adding stages can destroy a good system. Opus 4.6 sbs is degraded by five of six few-shot settings on hard rows (0.59 to 0.38), by structured outputs on medium rows (0.44 to 0.30), and by MJD function calling on hard rows (0.53 to 0.46, in Table G.1's no-self-correction setting) — the same shape as our result: the Opus-5 advocate alone is the best ranker (recall@5%FPR 33.3%, AUC 0.764, versus incumbent 9.5%), while the full critic stack collapses the ranking (AUC 0.468, ΔAUC −0.301). Our verdict was architectural: the critic stack supplies the FPR discipline the advocate lacks, so it became a veto/certification layer — the deployed D-rule-only veto at FPR@(A or B) = 2.0% — not a deletion. A stage must be scored on the axis it serves, not the headline metric.

(3) Cost as a result. One Opus sbs run over 52 queries ($2.87) buys about ten LensJudge `panel` gradings; our whole programme to date is ∼$244, in which `panel` ($0.27/candidate) and `multi-agent` ($0.33) against `lean` ($0.06) bought nothing (AUC 0.42 and 0.38 versus 0.51).

Adopt: their per-stage cost split, and the module ablation they should have run — measure each LensJudge stage's marginal contribution with a permutation test, but compute the null over *items*, not runs: our headline (recall@5%FPR 33.3% vs 9.5%, McNemar p 0.046) rests on ∼86 positives and carries the same small-n fragility.

Where it does not transfer: self-correction works because the database is an oracle: a bad query announces itself as a timeout or an undefined column, which is why all but one non-timeout error was repaired. Lens grading has no executor: a wrong grade raises nothing, it returns a number. Our JWST regrade showed what happens without one: three "independent" refuters correlated at κ 0.46-0.63, a geometry gate passing 2.3%, 23 of 24 known lenses rejected. Their evaluation also has no negative class: every test query has a correct answer, so PM has no false-positive axis. Ours is the opposite — ∼86 COWLS/literature positives against 400 catalogue-purified negatives, lenses at ∼1 in 10^4 massive ellipticals, a rule specified as FPR@(A or B) = 2.0%, not accuracy.

Bottom line

This is a careful, useful engineering paper: a real astronomical T2S benchmark, a genuine decomposition-and-self-correction ablation, and the rare virtue of publishing the ablations that failed. Its central weakness is statistical: ten-query bins with variance estimated over near-deterministic runs cannot support the significance claims drawn from them, a single-annotator gold set leaves the ceiling unmeasured, and two of the four modules are never turned off — which is where the residual error most plausibly lives. All three are cheap to fix. For us it is a well-matched external data point on decomposition and cost, and a clean illustration of why self-correction works in text-to-SQL — the database is an oracle — and why we cannot borrow it: nothing in a lens cutout throws an exception when the agent calls a merger an arc.